

# overview

the module, the project, and where everything belongs · prof. dr. nicolas meseth

- 1 why this module
- 2 what you will learn
- 3 the project
- 4 experiment: you are the receiver
- 5 1 + 4 challenges
- 6 how we work
- 7 this semester

📊 quick poll

have you ever written a program?

yes, quite a lot

a little, at school or for fun

never



[lfi.uber.space/q](https://lfi.uber.space/q)

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[lifi.uber.space/q](https://lifi.uber.space/q)

📊 quick poll

how often do you use ai chats like ChatGPT?

every day

every week

now and then

never



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📊 quick poll

have you ever worked with an AI agent, like Claude Code, Codex or OpenCode?

yes

i have heard of them

no, never heard of them



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📊 quick poll

how well do you know what happens inside a computer?

very well

roughly

hardly

not at all



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📄 quick quiz

how many different values can one byte hold?

8

16

256

no idea



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📱 quick quiz

a photo on your phone: roughly how many numbers are stored for it?

a few hundred

a few thousand

tens of millions

no idea



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[lifi.uber.space/q](https://lifi.uber.space/q)

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📄 quick quiz

at the lowest level, what does a computer work with?

words

pictures and sounds

numbers: zeros and ones

no idea



[lifi.uber.space/q](https://lifi.uber.space/q)

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[lifi.uber.space/q](https://lifi.uber.space/q)

📄 quick quiz

you switch off your computer. what happens to what is in its ram?

it stays there

it is lost

no idea



[lifi.uber.space/q](https://lifi.uber.space/q)

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☐  
☒  
☐

[lifi.uber.space/q](https://lifi.uber.space/q)

part 1

# why this module

📊 quick poll

how important will it be for your career to understand AI  
and to work well with it?

essential

very important

somewhat important

not important



[lifi.uber.space/q](https://lifi.uber.space/q)

how important will it be for your career to understand AI  
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somewhat important



not important



[lifi.uber.space/q](https://lifi.uber.space/q)

why this module

ai writes code in seconds.

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it explains anything you ask, and solves most exercises you give it.

why this module

ai writes code in seconds.

it explains anything you ask, and solves most exercises you give it.

so why learn anything at all?

the same question, fifty years ago:

everyone has  
a calculator.  
why still  
learn maths?



why maths, if you have a calculator?

to write the problem down so that the calculator can solve it.

why maths, if you have a calculator?

to write the problem down so that the calculator can solve it.  
to notice when 4,000 comes out and 40 would be right.

why maths, if you have a calculator?

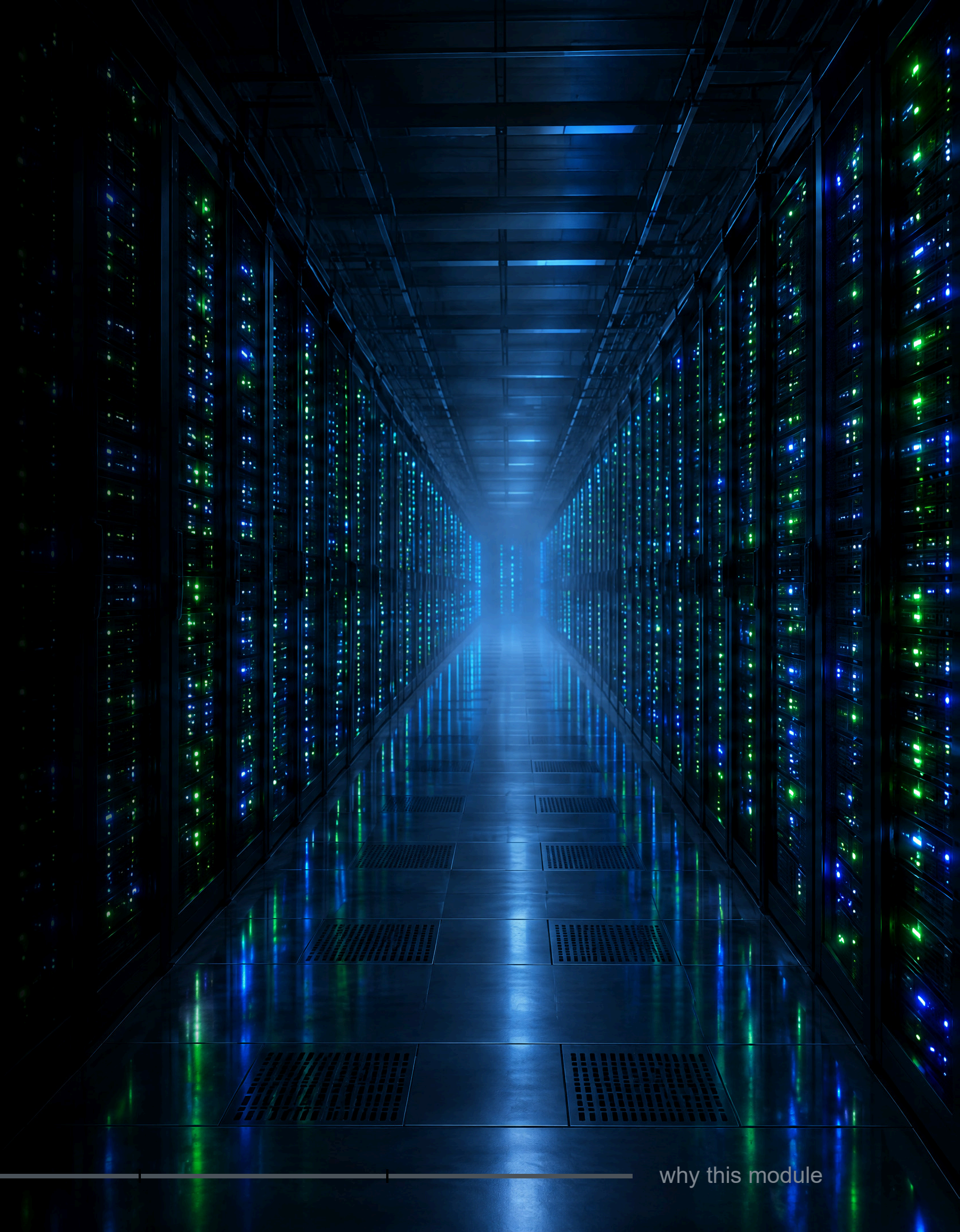
to write the problem down so that the calculator can solve it.  
to notice when 4,000 comes out and 40 would be right.  
to know what it cannot calculate at all.

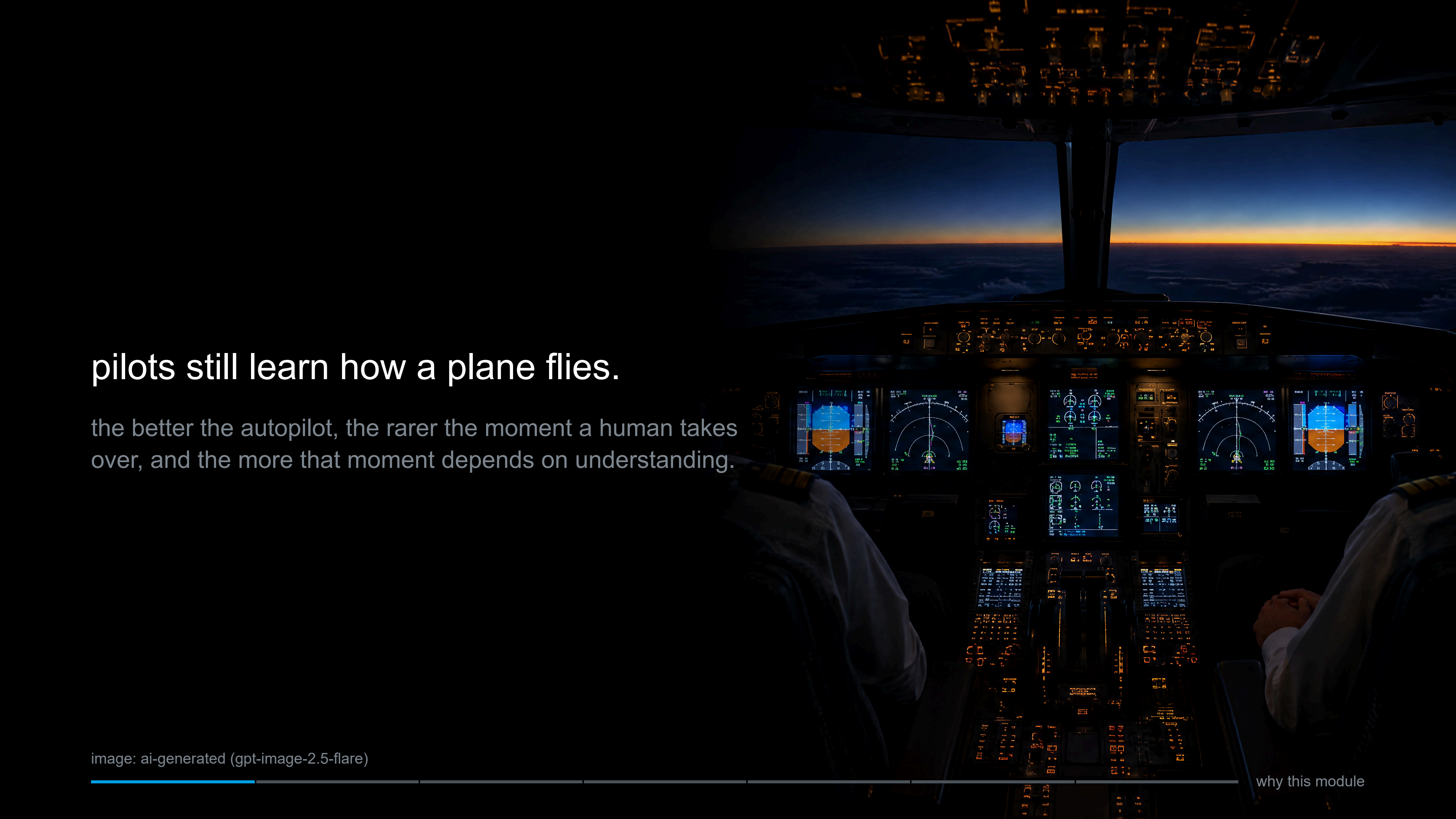
a computer is just a bigger calculator.

it turns numbers into numbers, billions of times a second. to  
judge what comes out, you have to understand what it does.

image: ai-generated (gpt-image-2.5-flare)

why this module





pilots still learn how a plane flies.

the better the autopilot, the rarer the moment a human takes over, and the more that moment depends on understanding.

the most powerful calculator humans built so far

it writes code in seconds.



image: ai-generated (gpt-image-2.5-flare, edited with gpt-image-2.5-sunburst)

why this module

the most powerful calculator humans built so far

it writes code in seconds.  
it sounds right, even when it is wrong.



image: ai-generated (gpt-image-2.5-flare, edited with gpt-image-2.5-sunburst)

why this module

the most powerful calculator humans built so far

it writes code in seconds.  
it sounds right, even when it is wrong.  
it cannot check its answer against the  
real world.



image: ai-generated (gpt-image-2.5-flare, edited with gpt-image-2.5-sunburst)

the most powerful calculator humans built so far

it writes code in seconds.  
it sounds right, even when it is wrong.  
it cannot check its answer against the  
real world.

you learn to work with it: cut, check,  
measure.



image: ai-generated (gpt-image-2.5-flare, edited with gpt-image-2.5-sunburst)

and to understand ai

ai

is software: a program, like any other.

and to understand ai

ai

is software: a program, like any other.

software

runs on a computer.

and to understand ai

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is software: a program, like any other.

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software

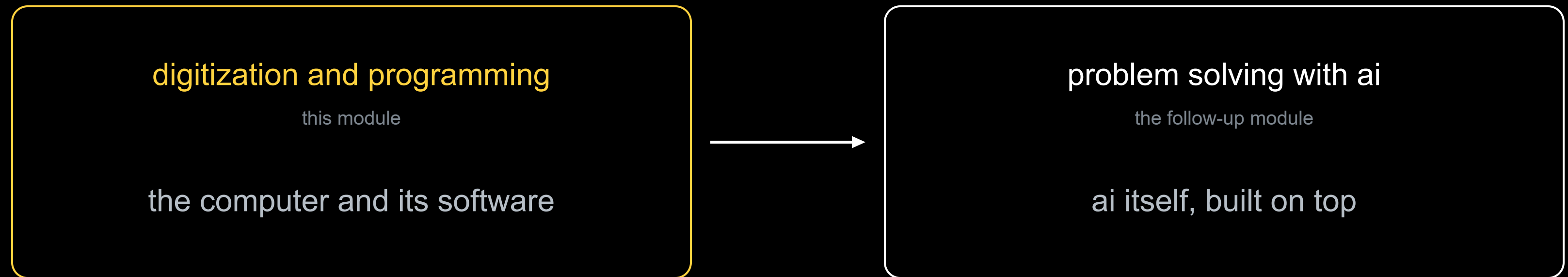
runs on a computer.

computer

turns numbers into numbers, billions of times a second.

to understand ai, you first understand the computer and its software.

what comes next



this module is the foundation. the next one builds on it.

part 2

# what you will learn

📊 quick poll

would you like to have a better understanding of computers?

yes, definitely

a little

not really



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what you will learn

how can we solve complex problems with computers?

## how can we solve complex problems with computers?

how do computers  
represent information?

how do computers  
store information?

how do computers  
transfer information?

how do computers  
process information?

## how can we solve complex problems with computers?

how do computers  
represent information?

making bits

how do computers  
store information?

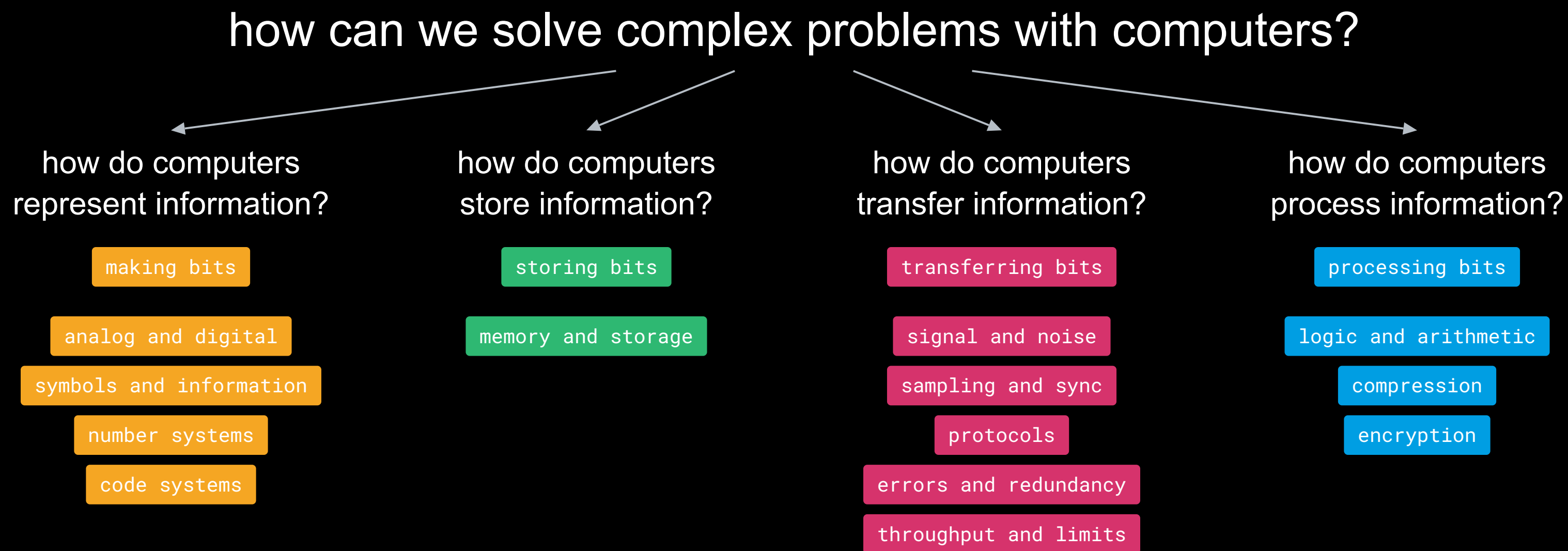
storing bits

how do computers  
transfer information?

transferring bits

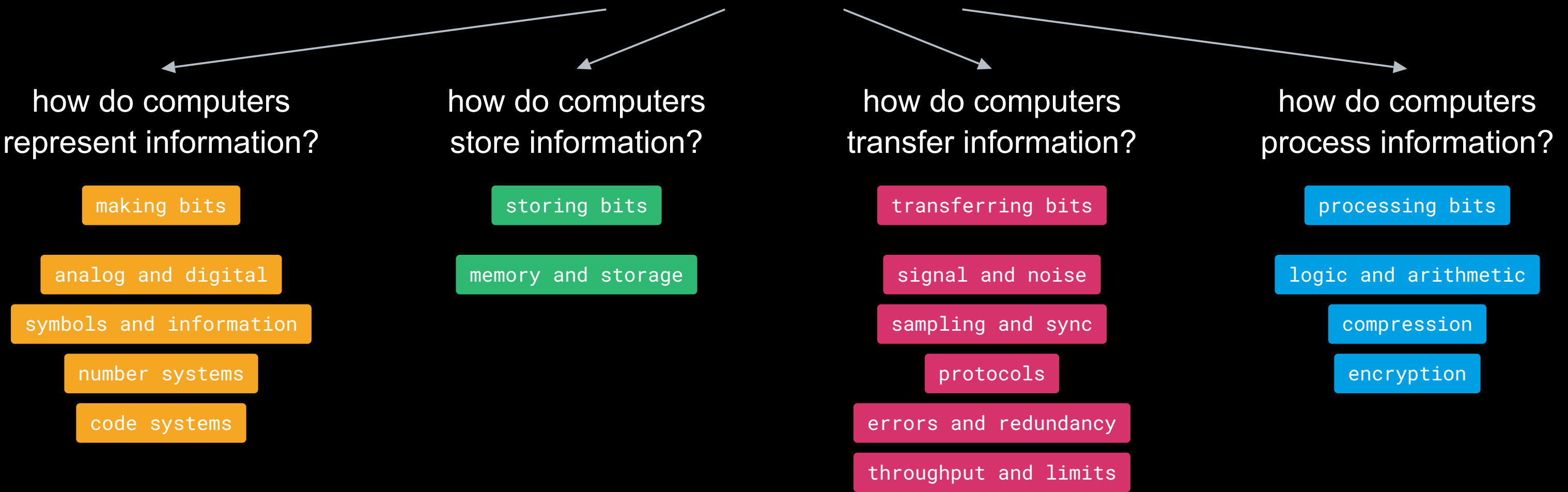
how do computers  
process information?

processing bits



- cutting problems
- input, processing, output
- algorithms and programs
- measuring and experimenting
- abstraction and layers

# how can we solve complex problems with computers?



the map lives on the website

LiFi Project

Concepts > Overview

Overview

The module, the project, and where everything belongs

This page is the map. It shows how the module is put together: the one question behind it, the four questions underneath, the eighteen concepts and their five colours, and the five challenges of the project with the concepts each of them needs. You will see this map again and again during the semester, on the slides and on this page, each time with the part lit up that we are working on. If you ever lose track of where you are, come back here.

One question, four questions

The module has one big question: **how can we solve complex problems with computers?** To answer it you need to understand what a computer does with information, and there are four things it does. It represents information, it stores it, it transfers it, and it processes it.

cutting problemsinput, processing, outputalgorithms and programsmeasuring and experimentingabstraction and layers

how can we solve complex problems with computers?

how do computers represent information?

making bitsanalog and digitalsymbols and informationnumber systemscode systems

how do computers store information?

storing bitsmemory and storage

how do computers transfer information?

transferring bitssignal and noisesampling and syncprotocolserrors and redundancythroughput and limits

how do computers process information?

processing bitslogic and arithmeticcompressionencryption

the lifi project

what you will learn

part 3

# the project

why a project,  
and not just  
lectures?



image: ai-generated from photos of the real device (gpt-image-2.5-sunburst)

the project

why a project?



## learn by doing

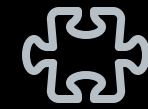
concepts stick when you have  
built them, broken them and  
fixed them yourself.

why a project?



## learn by doing

concepts stick when you have  
built them, broken them and  
fixed them yourself.



## problems first

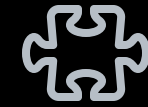
every problem meets you on  
your own device before it gets  
a name in class.

why a project?



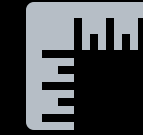
## learn by doing

concepts stick when you have built them, broken them and fixed them yourself.



## problems first

every problem meets you on your own device before it gets a name in class.



## reality answers

a device does not care how good an answer sounds. it works, or it does not.

the goal of the lifi project

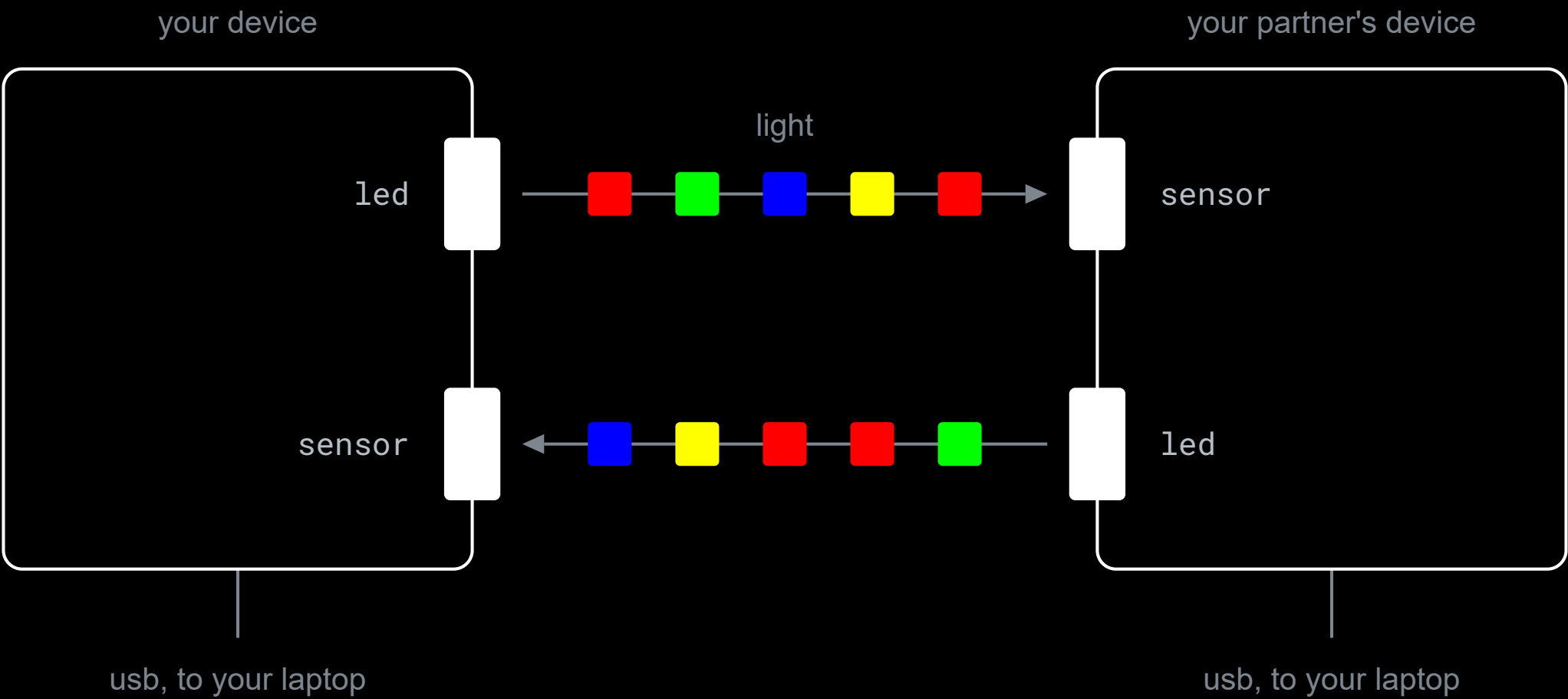
there is a photo  
on this laptop.  
how does it get  
to the other one?



image: ai-generated (gpt-image-2.5-flare)

the project

the device



a lamp and an eye on the front of each device. two devices face each other, so light travels both ways.

why a lamp

everything interesting in a computer is  
invisible.

this project makes it visible: a few times a second, on your  
desk, where you can explore it with your own hands.

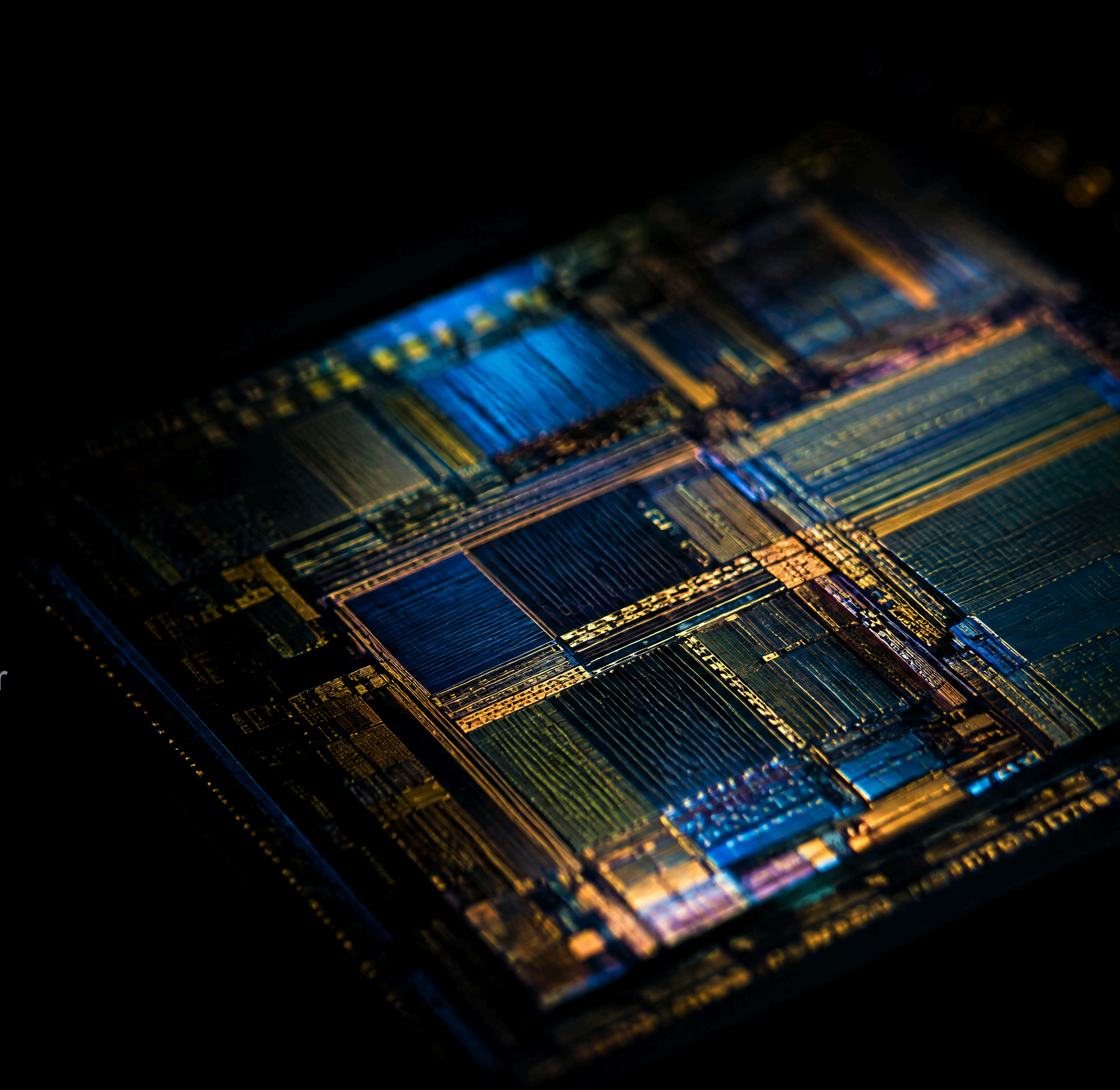


image: ai-generated (gpt-image-2.5-flare)

the project

the journey of one photo



a photo on your laptop

memory and storage

the journey of one photo



a photo on your laptop



becomes bytes

memory and storage

number systems

code systems

the journey of one photo



a photo on your laptop



becomes bytes



is packed into a frame with a checksum

memory and storage

number systems

code systems

protocols

errors and redundancy

compression

encryption

the journey of one photo



a photo on your laptop



becomes bytes



is packed into a frame with a checksum



bytes become colours

memory and storage

number systems

code systems

protocols

errors and redundancy

compression

encryption

symbols and information

analog and digital

the journey of one photo



a photo on your laptop



becomes bytes



is packed into a frame with a checksum



bytes become colours



travels as light

memory and storage

number systems

code systems

protocols

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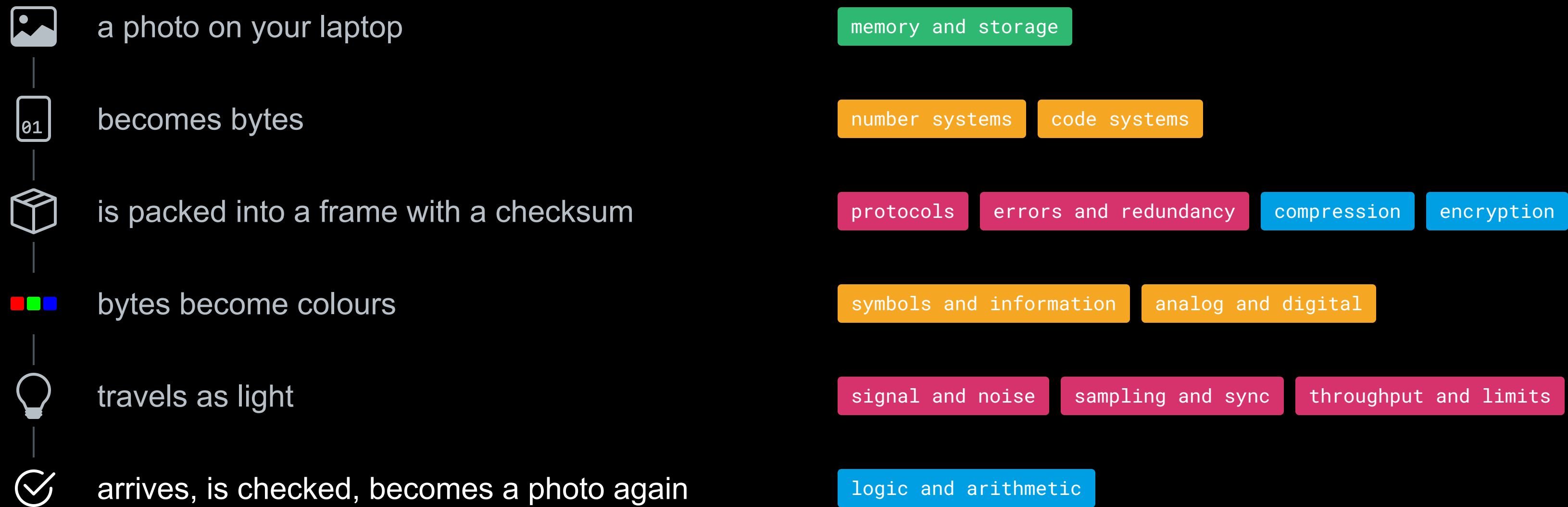
analog and digital

signal and noise

sampling and sync

throughput and limits

the journey of one photo



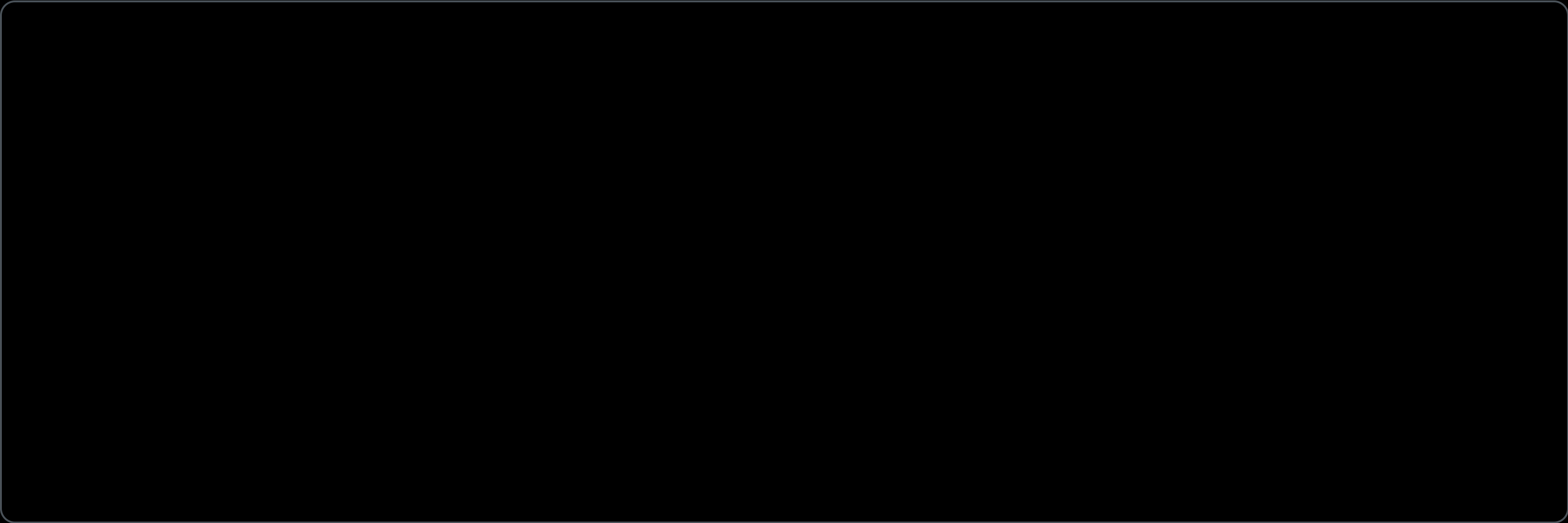
every stop on the way is a concept of this module.

part 4

experiment: you are the receiver

take a pen and a piece of paper.  
the screen is the lamp. you are the sensor.  
write down every symbol you see, using the alphabet at the bottom.

round 1: four colours, slowly



alphabet



r



g

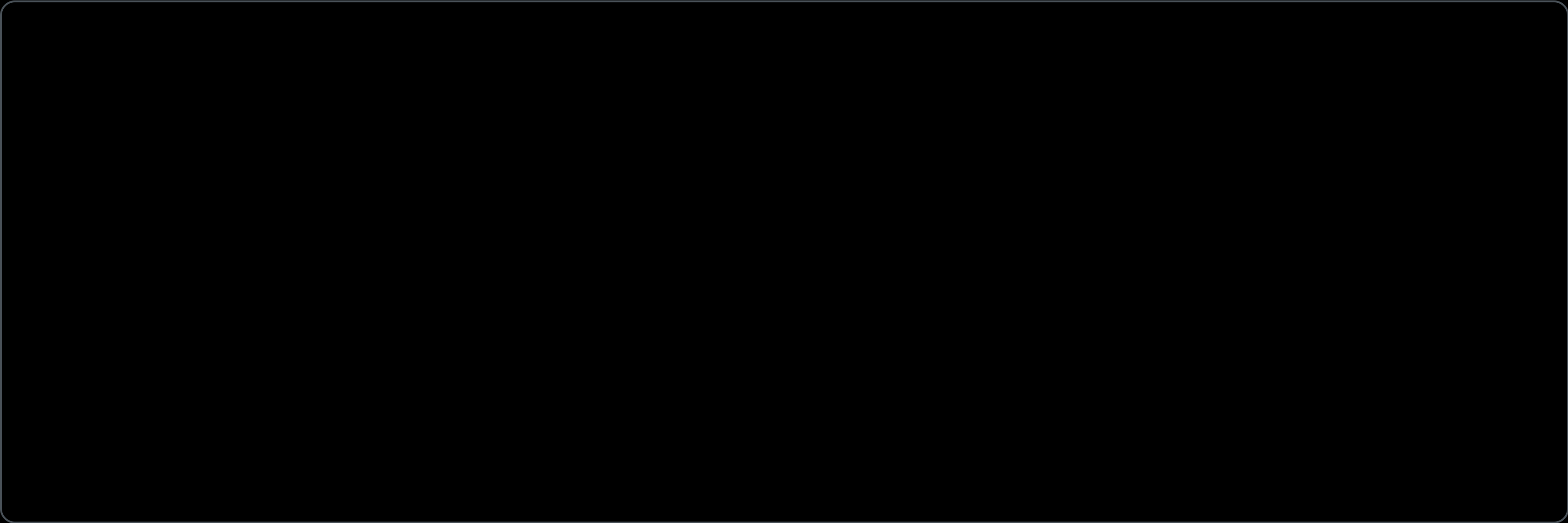


b



y

round 1: four colours, slowly



sent                    g   r   y   b   r   y   g   b

alphabet             r    g    b    y

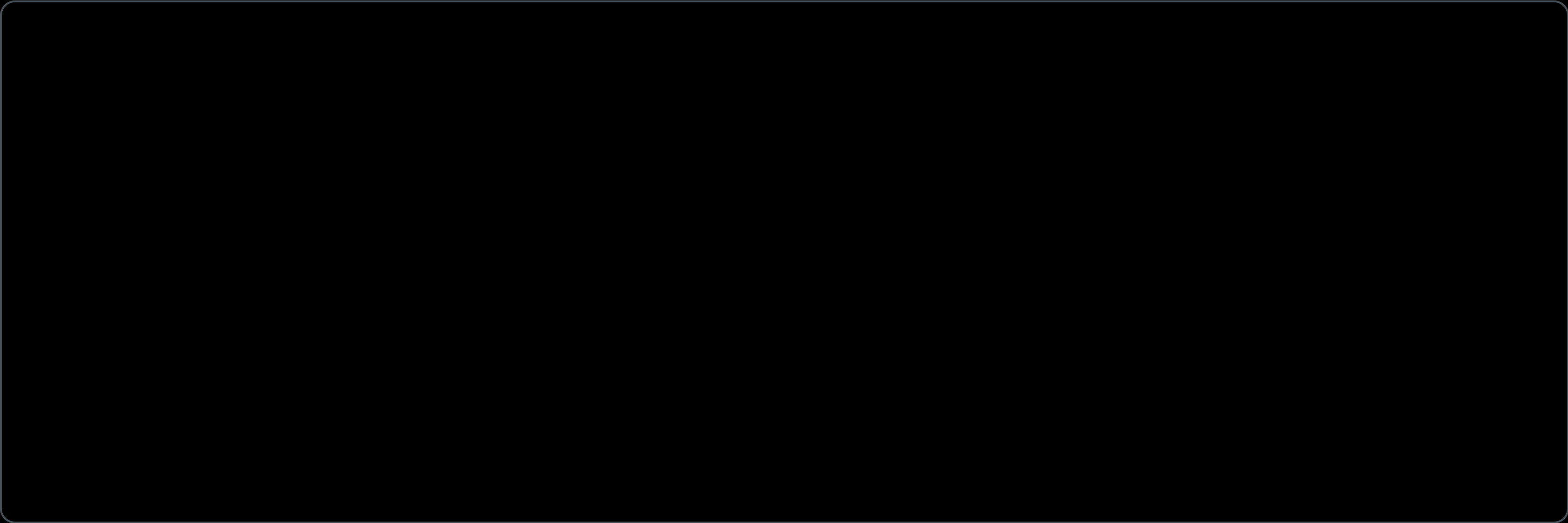
round 2: eight blues

alphabet

1 2 3 4 5 6 7 8

experiment: you are the receiver

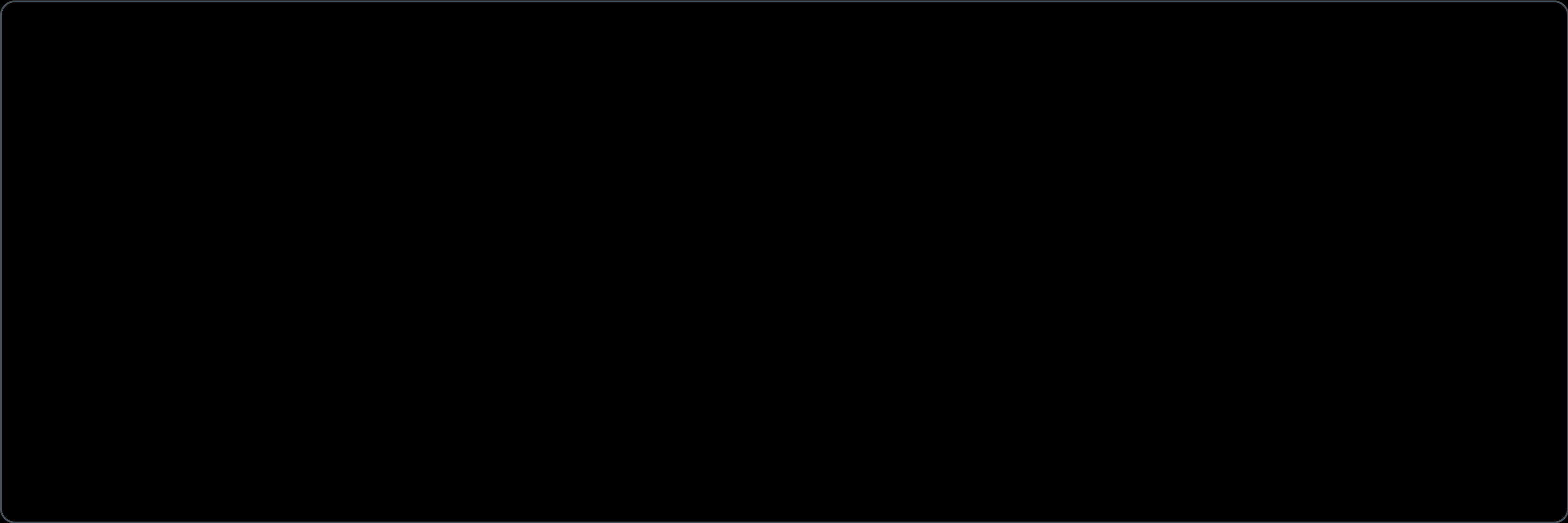
round 2: eight blues



sent            3   7   1   5   8   2   6   4

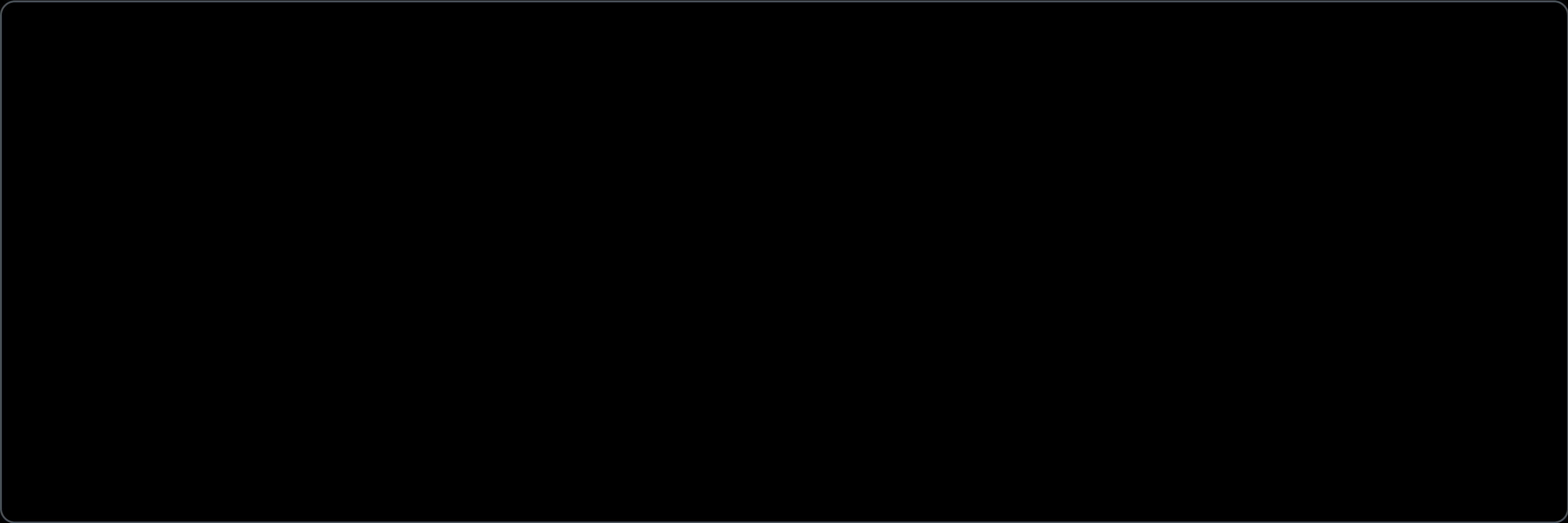
alphabet        1    2    3    4    5    6    7    8

round 3: faster



alphabet      ■ r    ■ g    ■ b    ■ y

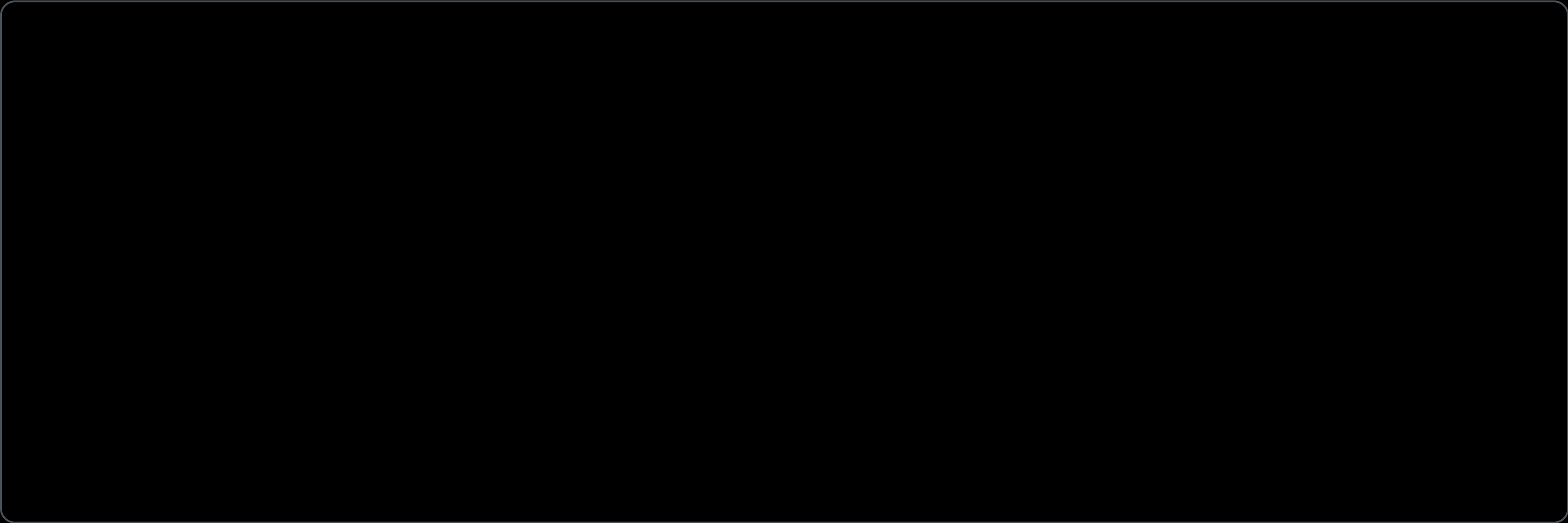
round 3: faster



sent            r   y   g   g   b   r   y   b   g   r   b   y

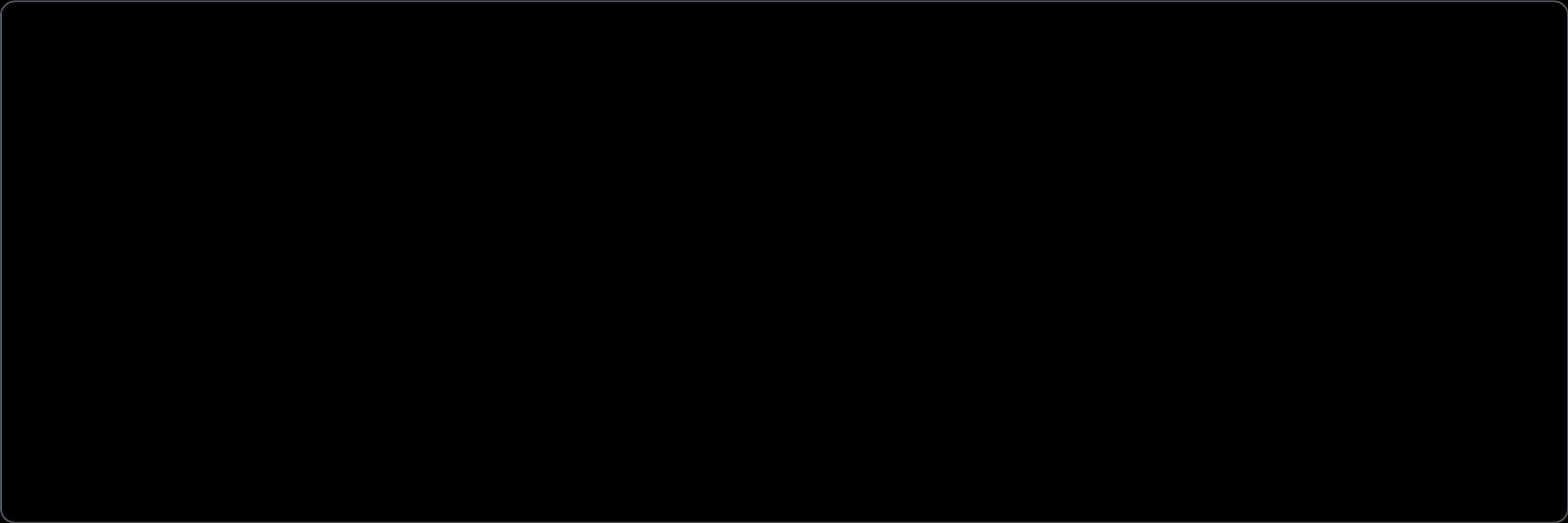
alphabet        r    g    b    y

round 4: where is the message?



alphabet       r     g     b     y

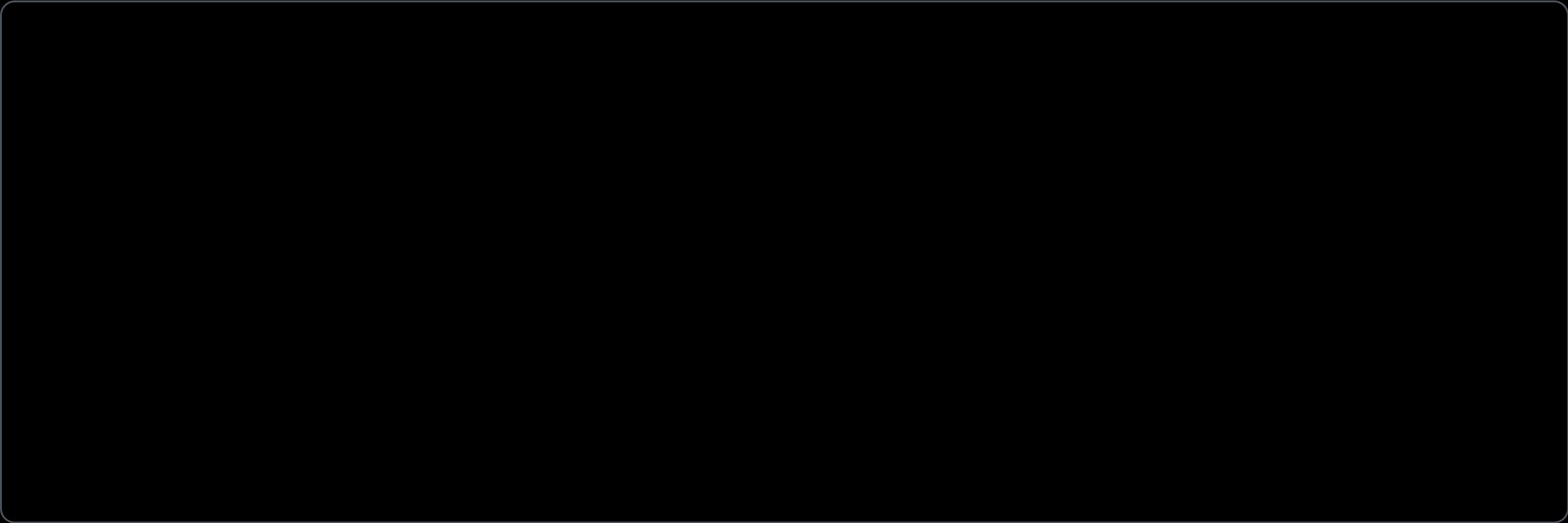
round 4: where is the message?



sent . . . . . g b r y g r . . . . .

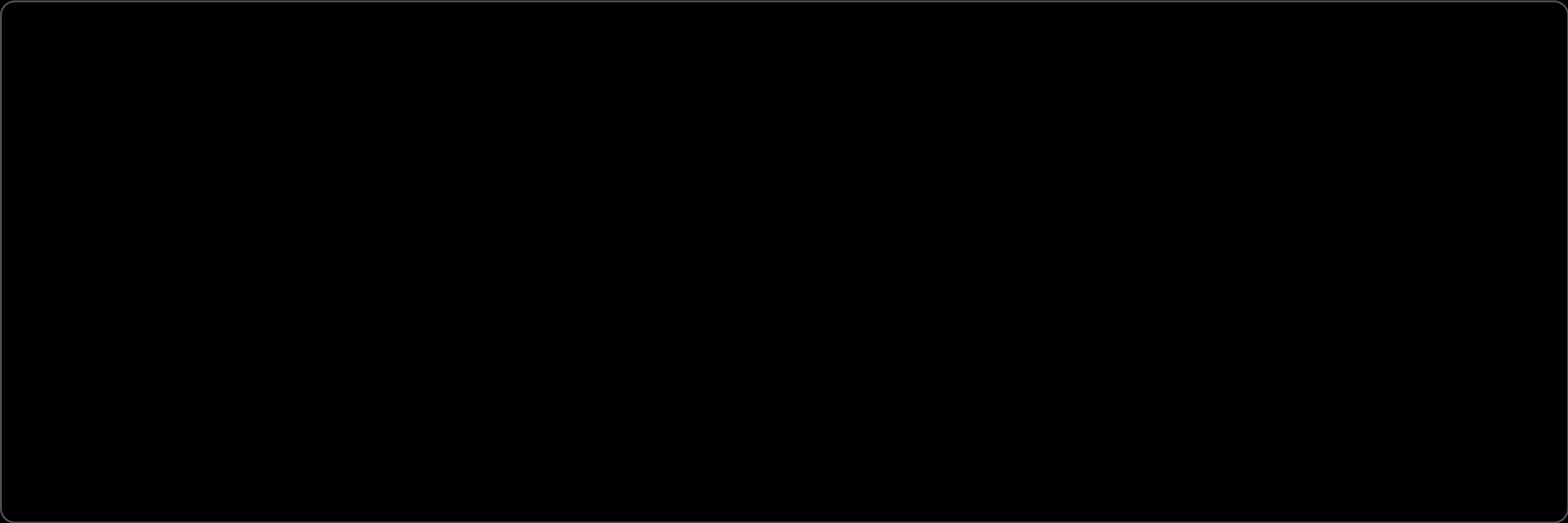
alphabet  r  g  b  y

round 5: did everything arrive?



alphabet      ■ r    ■ g    ■ b    ■ y

round 5: did everything arrive?



sent            y   g   r   b   g   y   b   r

alphabet        r    g    b    y





part 5

1 + 4 challenges

one new difficulty at a time

- |   |              |                                                |
|---|--------------|------------------------------------------------|
| 0 | the spark    | can you bring your device to life?             |
| 1 | the alphabet | how many signals can your receiver tell apart? |
| 2 | the word     | can you send a whole word, and how fast?       |
| 3 | the listener | can the receiver find the start on its own?    |
| 4 | the packet   | can you send a real file, correct and fast?    |

each challenge adds exactly one new difficulty. everything else stays as you know it.

what every challenge produces



## measurement logs

what did you change, what did you keep the same, and what came out?



## a description of your solution

enough for another team to rebuild it. from challenge 3 on, a full protocol specification.



## a log of mistakes

where the assistant suggested something, and a measurement proved it wrong.

nothing of this is handed in or graded. it is for your team.

part 6

# how we work



## two people, two devices

one device each, so you own  
both ends of the link. split the  
work, and swap now and  
then.



## two people, two devices

one device each, so you own  
both ends of the link. split the  
work, and swap now and  
then.



## the lamp has the last word

your assistant knows nothing  
about your device. measure  
before, measure after, and  
note where it was wrong.



## two people, two devices

one device each, so you own  
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work, and swap now and  
then.



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## try the demonstrators

a small toy in the browser for  
almost every concept: switch  
a byte, break a file, add noise.

part 7

this semester

## fourteen mondays

### 1 **kick-off: overview & setup**

devices, setup, challenge 0 starts

### 2 cutting problems, algorithms and programs

challenge 0: your first program

### 3 measuring and experimenting

challenge 0 check

### 4 analog and digital, symbols and information

challenge 1 starts

### 5 signal and noise

🏆 competition: challenge 1

### 6 number systems and code systems

challenge 2 starts

### 7 sampling and synchronization

challenge 2: faster

### 8 throughput and limits

🏆 competition: challenge 2

### 9 protocols

challenge 3 starts

### 10 logic and arithmetic

interop test

### 11 memory and storage

🏆 competition: challenge 3

### 12 errors and redundancy

standardisation session, challenge 4 starts

### 13 compression

challenge 4: faster and shorter

### 14 abstraction and layers

🏆 final: challenge 4

a typical monday, 15:00 to 18:15

15:00 look back: where are you stuck?

15:10 input: one concept

15:35 experiment, all together

16:05 workshop

16:55 break

17:10 workshop

18:00 your task for the week

more than half of every monday, you build.

every monday



**room HD0001**

mondays, 15:00 to 18:15.



**laptop and device**

bring both, every time. charge  
the laptop before you come.



**attendance list**

sign it every week. it is for  
statistics, not for grades.

the exam



**60 minutes**

multiple choice on the  
concepts, everyone on their  
own. the project is where you  
meet them.

the ranking in the competition does not count towards your grade.

the exam



**60 minutes**

multiple choice on the  
concepts, everyone on their  
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meet them.



**+5 % bonus**

if all five of your challenges  
are accepted. all or nothing.



the ranking in the competition does not count towards your grade.



**60 minutes**

multiple choice on the concepts, everyone on their own. the project is where you meet them.



**+5 % bonus**

if all five of your challenges are accepted. all or nothing.



**practise**

with the questions on every concept page. same format as the exam.

the ranking in the competition does not count towards your grade.

where to find what

|                                                                                     |                                                                                                     |                                                 |
|-------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------|-------------------------------------------------|
|    | <a href="https://docs.lifi-project.de">docs.lifi-project.de</a>                                     | everything to read: concepts, challenges, setup |
|    | <a href="https://lms.hs-osnabrueck.de">lms.hs-osnabrueck.de</a>                                     | announcements, changes, the schedule            |
|    | your course folder                                                                                  | your assistant and your code                    |
|  | <a href="https://lifi.uber.space/cockpit">lifi.uber.space/cockpit</a>                               | your link, live                                 |
|  | <a href="https://winf-hsos.github.io/lifi-concept-demos">winf-hsos.github.io/lifi-concept-demos</a> | the demonstrators                               |

📱 quick poll

how do you feel about this module?

looking forward to it

curious

a bit worried

worried



[lfi.uber.space/q](https://lfi.uber.space/q)

how do you feel about this module?

looking forward to it

curious

a bit worried

worried



[lfi.uber.space/q](https://lfi.uber.space/q)

by the end of term,  
a picture crosses that gap.  
and along the way,  
more than one light will go on.

